- Pipelining & Hazards Lec 11-15 } CA
- Lec 16 Exception } R

# LECTURE-13

Wednesday 14/10/26.

We can add a comparator in ID stage to see if the jump is required or not early. But This can't be done in RISC-V.

⇒ IF wrong prediction:

- Stop the pipeline & make control signals =0. Then flush the pipeline & move back to: beq 0x1, 0x2, 9

⇒ Dynamic Prediction: Has a table that maintains branch history that saves the target and your prediction.

PC

36: sub x10, x4, x8

40: beq x1, x3, 16 → for taking jump : 40 + (16 x 2)

*Multiply by 2 is done for sign extension & increasing a limit a bit.

Pearl Notes

Dynamic : (1 bit) → Changes current as previous ~~Bit~~ after each cycle.

Actual Results: NT, T, NT, T, T, NT, T, T, T, NT

first could be anything ← T, NT, T, NT, T, T, NT, T, T, T.

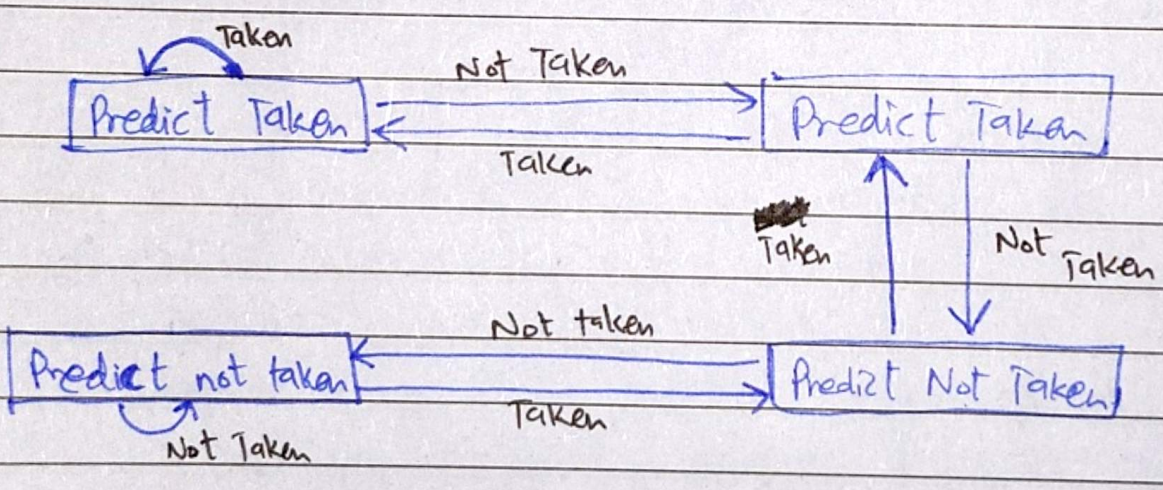
X X X X ✓ X X ✓ ✓ X

Correct = 3/10 (usually better)

Method:

Make the current prediction which was the ↑ actual result of previous cycle.

Dynamic (2 Bit): Change the decision only if 2 ~~Bit~~ consecutive NOT TAKEN.




☐ Exceptions: form of a Control Hazard.

SEPC: Supervisor exception program counter stores the PC of instruction jiski wajah say exception aie hai

SCAUSE: It includes the cause of exception

STVEC: Address of handler where the solution to handle int. lines supervisor trap vector

Handler: Includes address of location where how to handle is living. Control of that instr. is sent to handler. After it is handled; exception returned to same PC.